

Deep-Dive Explanation — D435f_UR_human_detect.py

Overview (What this node does)

This ROS 2 node fuses three real-time inputs:

- Color image → runs MediaPipe Hands to detect human wrists and YOLO to detect UR3 joints.
- Aligned depth-to-color → samples depth at the pixels of interest to estimate metric z.
- Camera intrinsics + TF2 → deprojects (u, v, z) to camera-frame 3D, then transforms to a base/world frame.

It then publishes 3D points and RViz markers and also emits an annotated debug image for visual verification.

Subscriptions, Publishers, Frames, QoS

Subscriptions (defaults):

- Color image: /camera/cam2/color/image_raw
- Aligned depth: /camera/cam2/aligned_depth_to_color/image_raw
- CameraInfo (color): /camera/cam2/color/camera_info

Frames:

- source_frame (camera optical): cam2_color_optical_frame
- target_frame (base/world): platform_base

Published topics:

- Human wrists: /wrist_left_point_base, /wrist_right_point_base (geometry_msgs/PointStamped)
- UR3 joints: /ur3/joint_markers (MarkerArray), /ur3/skeleton (Marker LINE_STRIP)
- Debug image: /wrist/debug_image (sensor_msgs/Image, annotated)

QoS: Subscribers use QoSProfile.SENSOR_DATA to match typical camera drivers; publishers use queue depth 10.

Key Parameters

- color_topic, depth_topic, info_topic: ROS topics for color, aligned depth, and CameraInfo.

- source_frame, target_frame: camera optical frame and base/world frame.
- min_det_conf, min_trk_conf: MediaPipe detection and tracking thresholds.
- yolo_model_path, yolo_conf, yolo_imgsz: YOLO model path, confidence threshold, and input size.
- ur3_class_names: CSV string listing UR3 joint class names in link order (e.g., base,shoulder,elbow1,elbow2,elbow3,wrist).

Human Wrist Pipeline (MediaPipe)

- 1) Convert color BGR → RGB and run MediaPipe Hands (max 2).
- 2) Take the wrist landmark (index 0) for each detected hand and convert normalized coords to pixel (u, v).
- 3) Robust depth at (u, v): take the median of non-zero pixels in 7×7, else try 11×11, else 15×15 windows.
- 4) Deproject (u, v, z) with fx, fy, cx, cy from CameraInfo to (Xc, Yc, Zc) in source_frame.
- 5) Transform to target_frame with TF2; publish PointStamped and a persistent sphere Marker (e.g., Left=red, Right=green).
- 6) Annotate the debug image with the label and z (meters).

UR3 Joints Pipeline (YOLO)

- 1) Run YOLO (Ultralytics) on the color frame.
- 2) For each expected class in ur3_class_names, keep the highest-confidence box; take the bbox center pixel.
- 3) Robust depth → deproject to camera 3D → TF2 to base/world; publish a sphere Marker per joint.
- 4) Build a LINE_STRIP skeleton through the available joints in order: base→shoulder→elbow1→elbow2→elbow3→wrist.
- 5) Draw bboxes/confidence values and a 2D skeleton on the debug image.

Depth, Alignment, and Dtype Handling

- Depth encodings supported: uint16 (millimeters, automatically divided by 1000) and float32 (meters).
- If an unexpected depth dtype is received, the sample is ignored and a warning is logged.
- If depth.header.frame_id differs from source_frame, a one-time warning suggests enabling depth-to-color alignment or adjusting source_frame.

TF Lookups

The node uses `lookup_transform(target, source, Time())` to get the latest transform, then `tf2_geometry_msgs.do_transform_point` to move points from camera to base/world. If TF is unavailable, it warns once and skips publishing for that frame.

Quick Setup Checklist

1. Start your camera with aligned depth enabled (e.g., RealSense `align_depth:=true`).
2. Ensure a TF chain exists from `target_frame` (parent) to `source_frame` (child) via URDF/`robot_state_publisher` or a static transform.
3. Confirm `CameraInfo` matches the color stream used for detection (same resolution/intrinsics).
4. Provide the correct YOLO weights and `ur3_class_names` order.

Troubleshooting Notes

- No `CameraInfo` or wrong intrinsics: verify the `.../camera_info` topic and that it matches the color stream.
- Depth zeros/NaNs: improve lighting, ensure the target lies within valid depth range; the adaptive window already attempts to fill holes.
- TF missing: publish a static TF or verify URDF; remember the transform direction is `target ← source`.
- UR3 classes mismatch: set `ur3_class_names` to your model's class names in link order.
- Performance: reduce YOLO `imgsz`, increase `yolo_conf`, or run with GPU acceleration.